

ZHEJIANG UNIVERSITY OF TECHNOLOGY

COMPUTER GRAPHICS

2025/2026(2)



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I. Story

The mission begins on a quiet battlefield. A tank rolls steadily forward across the terrain, its heavy tracks grinding through the ground. The camera watches from the side, tracking its slow advance.

Then the focus shifts. Across the field stands a rocket launcher — a sturdy tripod mounted with a sleek military rocket, armed and ready. Soldiers have set it up precisely for this moment.

The rocket ignites. Slowly at first, then with tremendous force, it tears away from the launcher and screams into the sky. It climbs upward, arcs through the air in a wide curve, accelerating hard as it locks onto its target.

The tank never stood a chance.

The rocket closes in at full speed — and then it hits. A massive explosion erupts at the point of impact. Fire rages where the tank once stood, thick flames consuming everything. The tank is gone. Nothing remains but the blaze burning against the battlefield sky.

II. My Computer settings

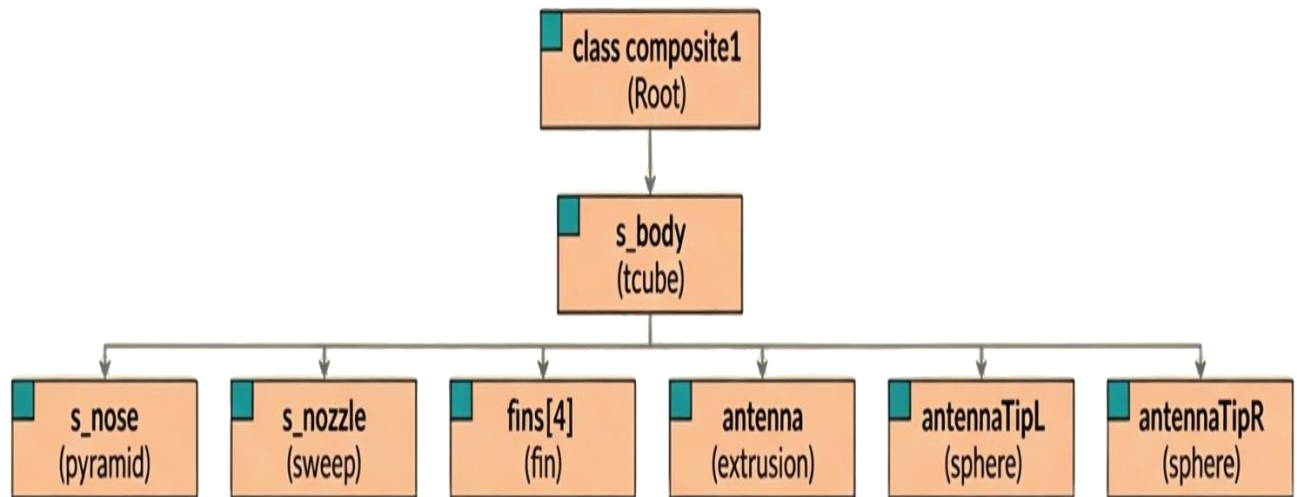
Device name	nn
Processor	12th Gen Intel(R) Core(TM) i7-1260P (2.10 GHz)
Installed RAM	16.0 GB (15.7 GB usable)
Device ID	2089E6A3-350B-433D-9C50-1B5680C529D6
Product ID	00331-20350-00000-AA180
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display

III. 3D Scene and Animation Characters

Animation Characters

Composite1	Military Rocket
Composite2	Military Tank
3D object	Rocket Launcher
2D object	Trees between objects.
2D animation object	Fire for TANK animation effect
Texture	On Battlefield to make it more real.

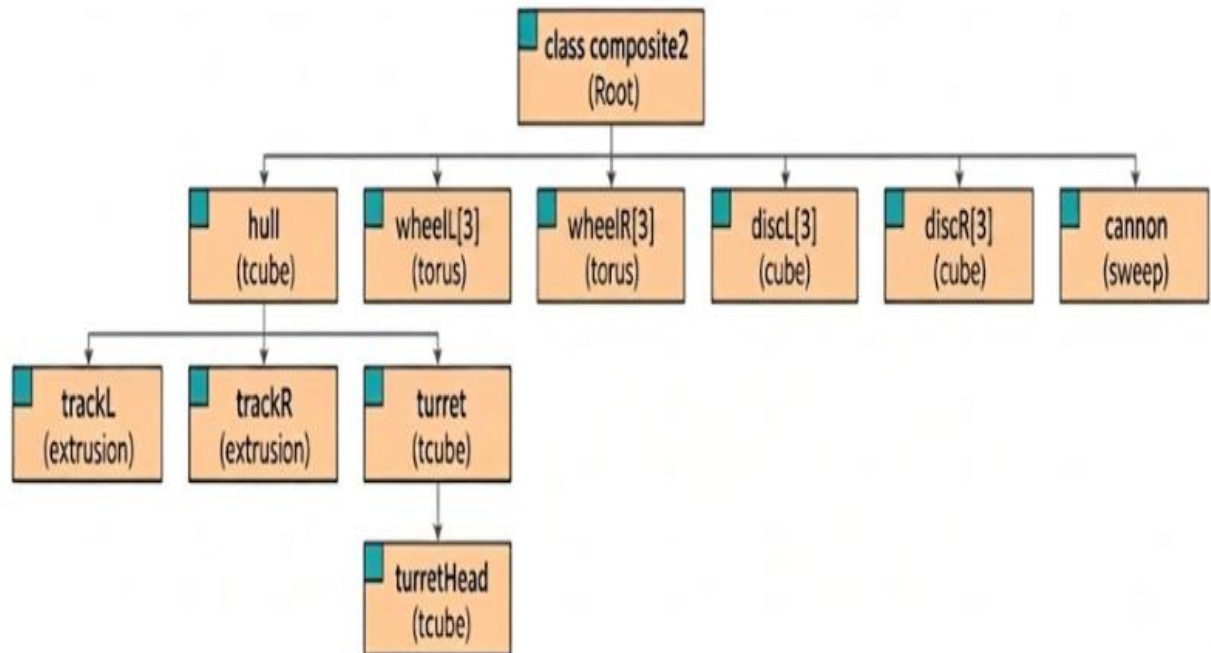
1. Hierarchical Structure of Composite1



2. Snap Shot of Composite1



3. Hierarchical Structure of Composite2



4. Snap Shot of Composite2



5. Other Important 3D models or Scene



IV. Conclusion and Suggestions for this course

This project gave me a much deeper understanding of how a real-time 3D scene is actually put together. Building the rocket and tank as composite objects made it clear why hierarchical structures matter so much: by attaching child parts such as the fins, the nozzle, the turret, and the wheels to a parent object with `setParent()`, I could move, rotate, and animate an entire assembly through a single transform instead of manually updating every individual shape. The wheel rotation was a particularly useful exercise, since the right-side wheels needed their angle negated to compensate for their mirrored orientation, which helped me understand how local transforms are inherited and combined down the hierarchy.

Working with sweep and extrusion objects also taught me the difference between profile-based and path-based geometry generation, and where each is appropriate. The rocket body and nozzle were a natural fit for sweep profiles, since they are surfaces of revolution, while the fins and tank tracks needed extrusion because they are not radially symmetric. I also learned some practical constraints along the way, such as sweep not supporting `makeInsideFaces()`, and that timing-based functions like `interpolate()` and `smoothAcceleration()` read the engine's internal frame time rather than accepting it as an argument, which changed how I structured the animation phases inside `animateForNextFrame()`.

For suggestions, I think the course would benefit from a short reference sheet summarising the engine's animation utility functions (`interpolate`, `smoothAcceleration`, `speedUp`) alongside small worked examples, since their exact parameter order and internal time-reading behaviour is not always obvious from the header files alone.

It would also help to cover audio integration earlier, since discovering that the engine requires WAV rather than MP3 files cost some time that could have been spent refining the animation itself. Overall, the assignment was a rewarding way to connect the theory of hierarchical transforms, particle systems, and lighting to a working, animated scene.